

Q1: Genetic Algorithm

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A logistics company is optimizing a drone delivery route between 5 locations:

D1, D2, D3, D4, D5

Distance matrix:

	D1	D2	D3	D4	D5
D1	0	9	7	14	10
D2	9	0	8	11	6
D3	7	8	0	5	12
D4	14	11	5	0	9
D5	10	6	12	9	0

Parents:

- P1: [D1, D3, D5, D2, D4]
- P2: [D4, D2, D1, D5, D3]

a. Crossover (Partially Mapped Crossover – PMX)

Use crossover points (2, 4). Generate two children.

b. Mutation

Apply inversion mutation between positions 2–4 on both children.

c. Fitness

Compute total distance for:

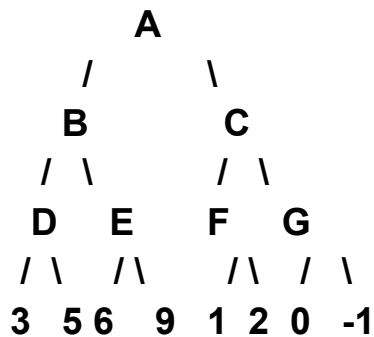
- Both parents
- Both mutated children

d. Selection Insight

Which chromosome is best and why?

Q2: Alpha Beta Pruning

Consider the following game tree (MAX at root):



Tasks:

- Apply Minimax and compute value of root
- Apply Alpha-Beta Pruning (left to right traversal)
 - Show pruned nodes
- Count:
 - Nodes visited
 - Nodes pruned